

4.1 Double Integrals over Rectangular Regions Worksheet

1. Find the volume of the solid beneath the paraboloid $f(x, y) = 12 - x^2 - 2y^2$ and above the region $R = \{(x, y) \mid 1 \leq x \leq 2, 0 \leq y \leq 1\}$.

2. Calculate $\int_0^1 \int_1^2 \frac{xe^{x^2}}{2y} dy dx$.

3. Determine the value of the volume communicated by the double integral $\iint_D 3 \, dA$ for each of the following regions, using only simple geometry formulas.

a) $D = \{(x, y) \mid 1 \leq x \leq 2 \text{ \& } 2 \leq y \leq 10\}$.

b) D is a circle of radius 4.

c) D is determined by the curves, $x = 0$, $y = 0$, & $y = 1 - x$.

4. Determine if each of the following statements are TRUE or FALSE.

a) Suppose that $f(x, y)$ is a continuous function over $\mathbb{R}^2$. There is no difference in the values of the integrals $\int_0^5 \int_{-1}^1 f(x, y) \, dx \, dy$ and $\int_{-1}^1 \int_0^5 f(x, y) \, dy \, dx$.

b) A double integral can be used to find volume under a surface and also the area of a region.

c) The double integral $\iint_D 3 \, dA$ represents triple the area of D .